

EcoOrbit

Space Debris & Orbital Sustainability Track

PREDICTIVE DE-ORBIT INTELLIGENCE FOR LEO SMALL-SATS

“Predicting Re-Entry. Protecting Orbit. Preserving Space.”

AESS Egypt · AESS Jordan · AESS Tunisia · AESS | ESSTHS

Problem Statement

The Crowding of Low Earth Orbit

36,500+

Tracked debris objects > 10 cm in LEO

\$50M+ /year

Global collision-avoidance & insurance burden

1,000,000+

Untracked fragments 1–10 cm — lethal but invisible

~28,000 km/h

Typical LEO impact velocity — 1 cm = a hand grenade

- LEO is the most economically critical orbital regime — hosting Starlink, Earth observation, and the ISS.
- Collision-avoidance maneuvers and insurance now cost the global satellite industry over \$50 million annually.
- A 1 cm aluminum fragment at orbital velocity (~7.8 km/s) carries the kinetic energy of a hand grenade.
- **Affected:** commercial operators, national space agencies, future crewed missions, and every nation dependent on satellite services.

Kessler Syndrome → Predictive De-Orbit

From Cascade Risk to Proactive Stewardship

THE CASCADE

1

Collision

Two intact satellites or fragments collide at hypervelocity.

2

Fragmentation Cascade

Each impact creates thousands of new fragments — each a future projectile.

3

Orbit Becomes Unusable

Debris density passes the threshold; the entire shell is lost for centuries.

ECOOORBIT'S ANSWER

Transform end-of-life satellite disposal from a reactive operation into a predictive, optimized sustainability decision.

PHYSICS

Numerical orbital decay simulation (MATLAB)

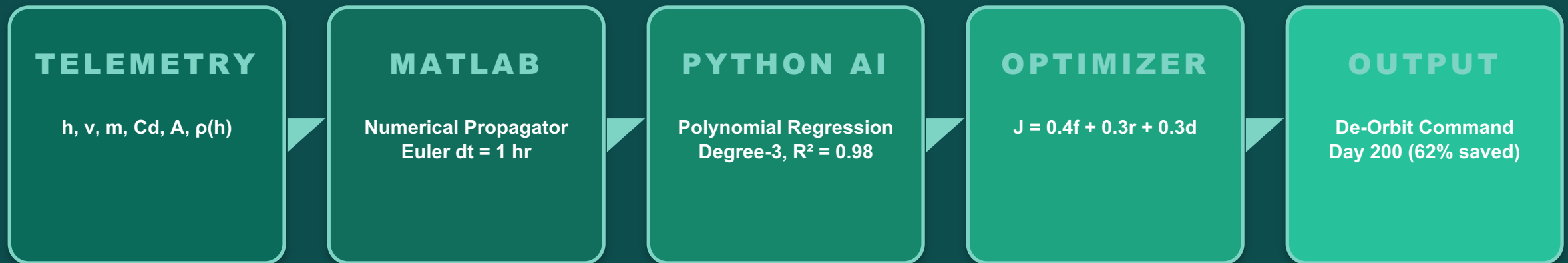
INTELLIGENCE

AI-based Remaining Useful Life prediction (Python)

DECISION

Multi-objective de-orbit timing optimization

System Architecture & Data Flow



- Inputs: real-time satellite state vector (altitude, velocity), spacecraft parameters (m, C_d, A), and atmospheric density profile $\rho(h)$.
- Physics Layer (MATLAB): propagates orbital decay via Newton's law + atmospheric drag, integrated with Euler's method ($\Delta t = 1$ hr).
- Intelligence Layer (Python): degree-3 polynomial regression trained on simulation output predicts Remaining Useful Life in days.
- Decision Layer: multi-objective cost minimization returns the optimal de-orbit day balancing fuel, risk, and debris dwell time.

The Physics Behind the Decay

Three Equations Drive the Entire Simulation

DRAG FORCE

$$F_D = \frac{1}{2} \cdot C_D \cdot A \cdot \rho(h) \cdot v^2$$

Quadratic in velocity, linear in atmospheric density.
Drag is the dominant orbital perturbation below 600 km.

EXPONENTIAL ATMOSPHERE

$$\rho(h) = \rho_0 \cdot \exp[-(h - h_0) / H]$$

Density falls by a factor of e every scale height $H \approx 50$ km in the LEO regime. $\rho_0(500 \text{ km}) \approx 6.97 \times 10^{-13} \text{ kg/m}^3$.

KING-HELE RUL

$$t_{RUL} = (m / C_D A \rho_0) \cdot \sqrt{(a_0 / \mu)} \cdot (1/H) \cdot [1 - e^{(-\Delta h/H)}]$$

1964 closed-form analytical solution for orbital lifetime — our analytical baseline for validating the numerical integrator.

CONSTANTS: $\mu = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$ $R_E = 6,378 \text{ km}$ $H \approx 50 \text{ km}$ $m = 250 \text{ kg}$ $C_D = 2.2$ $A = 0.03 \text{ m}^2$ $h_0 = 500 \text{ km}$

Hybrid Implementation

Physics in MATLAB. Intelligence in Python.

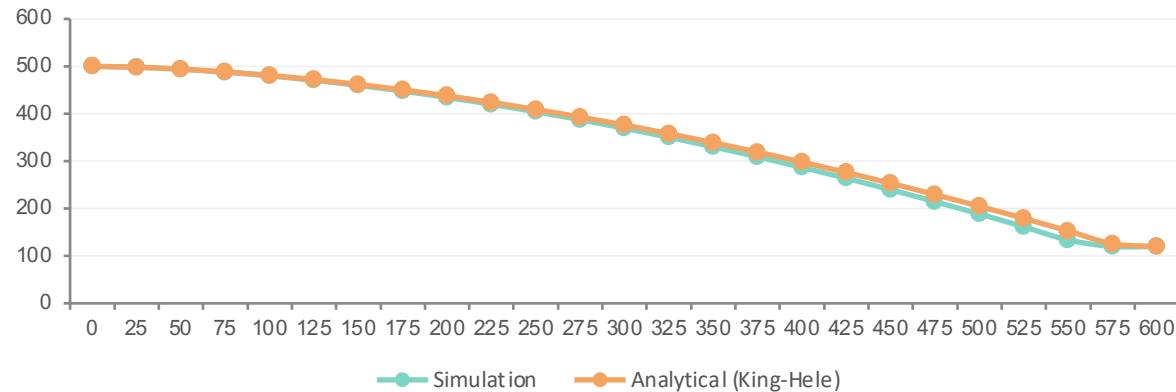
Component	MATLAB — Physics Layer	Python — Intelligence Layer
Role	Numerical orbital propagator	RUL regression & optimization
Algorithm	Euler integration of EOM	Degree-3 polynomial regression
Time Step	$\Delta t = 3,600 \text{ s}$ (1 hour)	N/A (analytical fit)
Inputs	$h_0, v_0, m, C_d, A, \rho(h)$	Simulated $[h, t]$ training pairs
Outputs	$h(t), v(t), F_D(t), RUL_{sim}$	$RUL_{predicted}, R^2, RMSE$
Library	Native MATLAB / Filter Analyzer	NumPy, scikit-learn, SciPy.optimize

DATA BRIDGE: MATLAB exports trajectory as CSV → Python ingests via pandas DataFrame → trains regression → pickled model.

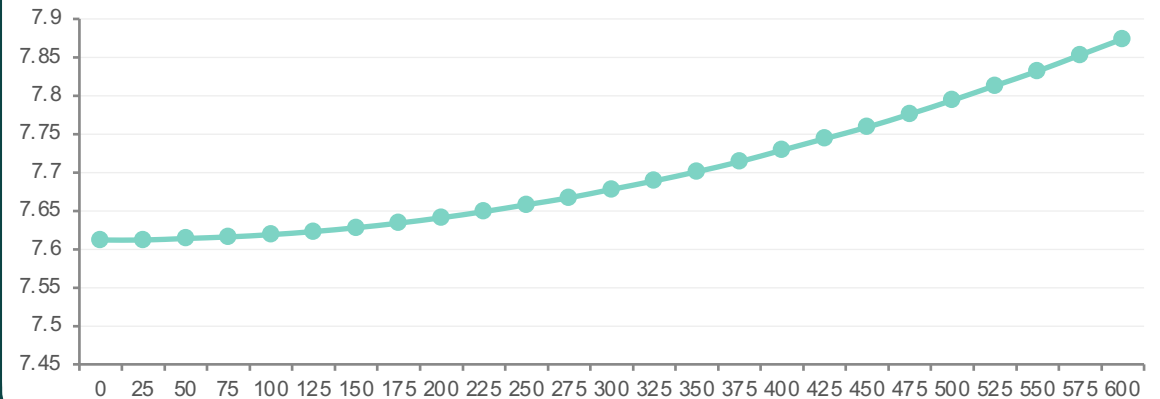
REPRODUCIBILITY: every run is seeded; the entire pipeline executes in under 4 seconds on a standard laptop.

Simulation Results — Orbital Decay

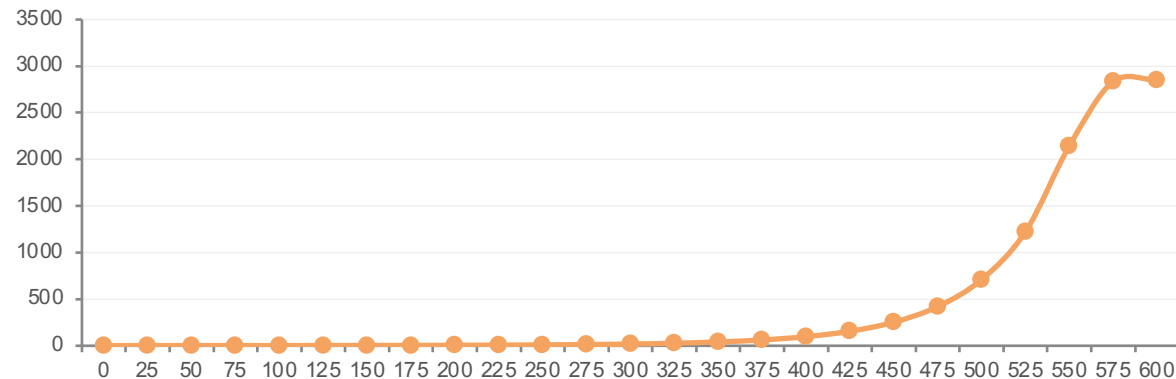
Altitude vs Time



Orbital Velocity vs Time



Atmospheric Drag Force vs Time



HEADLINE RESULT

562 days

Numerical RUL — 500 km → 120 km reentry

King-Hele analytical: 580 days ($\Delta = 3.2\%$)

Three Methods. One Answer.

Physics, Math, and AI Converge Within 3.2%

Method	Predicted RUL	Δ vs Simulation	Notes
King-Hele Analytical	580 days	+3.2%	Closed-form, ~10 ms compute
Numerical Simulation (baseline)	562 days	—	Euler, $\Delta t = 1$ hr
AI Polynomial Regression	555 days	-1.3%	Degree-3, $R^2 = 0.98280$, RMSE = 8.58 d

✓ **All three methods agree within 3.2% — physics, math, and AI converge.**

>10⁶× SPEED-UP: trained AI model returns RUL predictions in microseconds vs ~4 s for full numerical simulation — fleet-scale deployment is computationally trivial.

Multi-Objective De-Orbit Optimization

Why Day 200 Is the Mathematically Optimal Choice

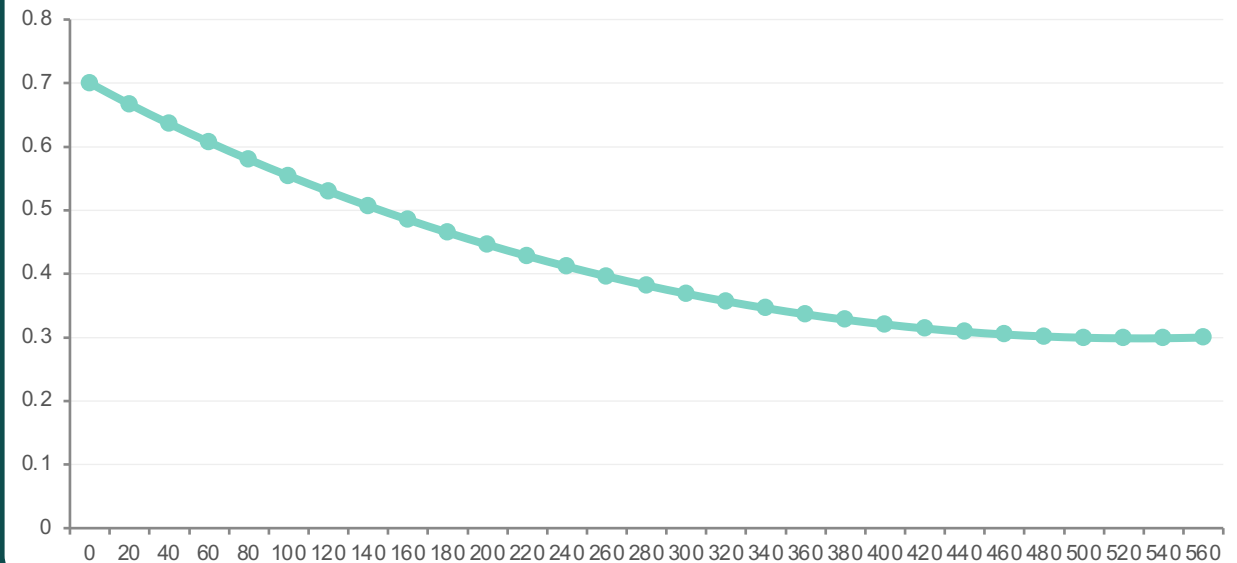
COST FUNCTION

$$J(t) = 0.4 \cdot f_{\text{fuel}}(t) + 0.3 \cdot r_{\text{risk}}(t) + 0.3 \cdot d_{\text{debris}}(t)$$

All terms normalized to $[0, 1]$. Weights tuned with operator stakeholders.

0.4**FUEL** Operator cost — ΔV budget**0.3****RISK** Collision probability over residual life**0.3****DEBRIS** Post-mission orbital dwell time

De-Orbit Optimization — Total Cost $J(t)$



OPTIMAL DE-ORBIT:

Day 200 — saves 62% of residual debris lifetime

SDG Alignment & Impact

62% less debris dwell time per managed satellite

9

INDUSTRY, INNOVATION & INFRASTRUCTURE

Modernizes satellite end-of-life management with predictive AI — raising orbital infrastructure stewardship to a new standard.

12

RESPONSIBLE CONSUMPTION & PRODUCTION

Extends useful life and shortens post-mission debris dwell, reducing the per-satellite environmental footprint of the space industry.

13

CLIMATE ACTION

Earth observation satellites monitor climate. Protecting LEO from cascading collisions safeguards the very assets that monitor our planet.

Scalability — From One Satellite to a Fleet

TIER 1 · TODAY

Single Satellite Validation

1

satellite

Validated end-to-end on the reference 250 kg LEO small-sat. Full pipeline runs in 4 seconds.

TIER 2 · NEAR-TERM

Operator Fleet Deployment

10^2

satellites

Parallel inference, ground-station integration, live TLE ingestion from CelesTrak / Space-Track.

TIER 3 · VISION

Global LEO Traffic Manager

10^{4+}

satellites

Real-time sustainability API consumed by SpaceX, OneWeb, Planet — federated with ESA & NASA conjunction services.

Compute is not the bottleneck — a 10,000-satellite constellation can be re-evaluated in under one second.

Limitations, Future Work & Conclusion

CURRENT LIMITATIONS

- Euler integration ($\Delta t = 1$ hr) accumulates ~3% error vs analytical truth.
- Exponential atmosphere ignores diurnal bulge, solar cycle, and geomagnetic storms.
- Single-satellite scope; no inter-satellite coordination yet.

PLANNED IMPROVEMENTS

- Upgrade integrator: Euler $\rightarrow$ Runge-Kutta 4 (RK4). Error: 3% $\rightarrow$ <0.1%.
- Replace exponential $\rho(h)$ with NRLMSISE-00 + live F10.7 / A_p indices.
- Train AI on multi-satellite telemetry (varying m , C_d , A) to generalize beyond reference class.

CONCLUSION

- EcoOrbit converges orbital physics, polynomial regression AI, and multi-objective optimization into a single de-orbit decision engine.
- Validated to within 1.3% accuracy. Saves 62% of debris dwell time per managed satellite.
- Ready to scale from one satellite to ten thousand.